A

Major Project Report

On

Fake News Classification Using ML Models

Submitted to Jawaharlal Nehru Technological University, Hyderabad, in partial fulfillment of the requirements for the award of the degree in

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted By

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Under the Guidance of

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Assistant Professor

ANUBOSE INSTITUTE OF TECHNOLOGY FOR WOMENS

UGC Autonomous Institution

Approved by AICTE New Delhi, Affiliated to JNTUH. Accredited by NAAC with "B++" Grade

New Paloncha, Bhadradri Kothagudem, Telangana

2025-26

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New Paloncha, Bhadradri Kothagudem, Telangana

CERTIFICATE

This is to certify that the project report entitled **"Fake News Classification Using ML Models"** being submitted by

Bhuky Bhoomika -22PP1A0505

in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, Jawaharlal Nehru Technological University Hyderabad, is a record of Bonafide work carried out under my guidance and supervision. The results embodied in this project report have not been submitted to any other University or institute for the award of any Degree.

Internal Guide

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Assistant Professor

Head of the Department

Dr. D. Praneeth Ph.D

Assistant Professor

External Examiner

Principal

DECLARATION

I declare that this project report "**Fake News Classification Using ML Models"** submitted in partial fulfillment of the degree of B. Tech in Computer Science and Engineering is a record of original work carried out by us under the guidance of**. Dr.D.Praneeth Ph.D,** Assistant Professor and was not submitted to any other Institution or University for award of any other degree.

Project Associates:
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I express our deep sense of gratitude and thanks to our project guide **Dr.D.Praneeth Ph.D,Assistant Professor** of Computer Science and Engineering under whose guidance and supervision this work had been accomplished.

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ABSTRACT

The rapid growth of digital media and social networking platforms has significantly increased the spread of fake news, posing serious threats to information credibility, public trust, and social stability. This project presents a robust and automated Fake News Detection System that leverages machine learning and natural language processing (NLP) techniques to accurately classify news articles as either real or fake.

The system utilizes a data-driven approach, where news datasets are collected and preprocessed to remove noise, stopwords, and irrelevant textual elements. Feature extraction is performed using the Term Frequency–Inverse Document Frequency (TF-IDF) vectorization technique, which converts textual data into meaningful numerical representations suitable for machine learning models.

To enhance classification accuracy and reliability, the system employs multiple supervised learning algorithms, including Logistic Regression, Random Forest Classifier, and Extreme Gradient Boosting (XGBoost). Logistic Regression is used for its efficiency and effectiveness in binary classification tasks, Random Forest improves performance through ensemble learning and decision tree aggregation, and XGBoost provides high accuracy and optimization through gradient boosting techniques. These models are trained and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score.

A Flask-based web application is developed to provide a user-friendly interface, allowing users to input news articles and receive real-time predictions along with probability scores from each model. The system also incorporates a majority voting mechanism to determine the final classification, thereby improving robustness and reducing model bias.

Overall, this project demonstrates the practical application of machine learning algorithms in combating misinformation.

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Tables and Abbreviations

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Abbreviations:

AI : Artificial Intelligence ANN:
Artificial Neural Network
API : Application Programming Interface
CNN: Convolutional Neural Network CSV:
Comma Separated Values
CSS: Cascading Style Sheets
DB: Database
DFD: Data Flow Diagram
HTML: Hyper Text Markup Language IDE:
Integrated Development Environment JS:
JavaScript
KNN: K-Nearest Neighbors
ML: Machine Learning MVT:
Model View Template
NLP: Natural Language Processing OS:
Operating System
RNN: Recurrent Neural Network
SVM: Support Vector Machine SQL:
Structured Query Language UML:
Unified Modeling Language UAT:
User Acceptance Testing UI: User
Interface
URL: Uniform Resource Locator

CHAPTER – 1

1. INTRODUCTION

The advancement of information technology and the widespread adoption of the internet have revolutionized the way people access, consume, and share information. In today's digital age, news is disseminated rapidly through online platforms such as social media, blogs, and news websites. While this transformation has made information more accessible and real-time, it has also led to a significant challenge—the rapid spread of fake news. Fake news refers to false, misleading, or fabricated information that is presented as legitimate news, often with the intent to deceive readers or manipulate public opinion.

The proliferation of fake news has become a global concern, impacting various aspects of society including politics, healthcare, and social harmony. During critical events such as elections, pandemics, or natural disasters, the spread of misinformation can lead to confusion, panic, and harmful consequences. Traditional methods of verifying news, such as manual fact-checking by experts, are no longer sufficient due to the enormous volume of information generated daily. These methods are time-consuming, labor-intensive, and unable to keep pace with the speed at which fake news spreads online.

To address this issue, there is a growing need for automated systems that can efficiently and accurately detect fake news. Machine learning (ML) and natural language processing (NLP) have emerged as powerful tools in tackling this problem. These technologies enable systems to analyze textual content, identify patterns, and classify information based on learned features. By leveraging these techniques, it is possible to develop intelligent systems that can distinguish between real and fake news with high accuracy.

This project focuses on the development of a Fake News Detection System using machine learning algorithms and NLP techniques. The system is designed to analyze news articles and predict their authenticity in real time. The first step in the process involves collecting and preprocessing a dataset containing labeled news articles. Data preprocessing includes cleaning the text by removing stopwords, punctuation, and irrelevant characters, as well as converting

text into a standardized format. This step ensures that the data is suitable for further analysis and model training.

Feature extraction is performed using the Term Frequency–Inverse Document Frequency (TF-IDF) technique, which transforms textual data into numerical vectors. TF-IDF helps in identifying important words in a document by considering both their frequency within the document and their rarity across the entire dataset. This representation plays a crucial role in improving the performance of machine learning models.

The system employs multiple supervised learning algorithms to classify news articles, including Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost). Logistic Regression is widely used for binary classification problems and provides a strong baseline model. Random Forest, an ensemble learning method, combines multiple decision trees to improve prediction accuracy and reduce overfitting. XGBoost, a powerful gradient boosting algorithm, further enhances performance by optimizing model learning through iterative improvements. By using multiple models, the system ensures better reliability and robustness in predictions.

To make the system accessible and user-friendly, a web-based interface is developed using the Flask framework. This interface allows users to register, log in, and input news articles for prediction. The system processes the input and provides the classification result along with confidence scores from each model. Additionally, a majority voting mechanism is implemented to determine the final output based on the predictions of all models, thereby increasing overall accuracy.

The Fake News Detection System not only provides an effective solution for identifying misinformation but also contributes to raising awareness about the importance of verifying news sources. It demonstrates how modern technologies can be applied to solve real-world problems in an efficient and scalable manner. Furthermore, the modular design of the system allows for future enhancements, such as incorporating deep learning models, expanding to multilingual datasets, and integrating real-time fact-checking APIs.

CHAPTER – 2

2. LITERATURE REVIEW

The problem of fake news detection has gained significant attention in recent years due to the rapid growth of online media platforms. Researchers have explored various techniques from data mining, machine learning, and natural language processing to identify and classify misleading information. Early approaches focused on manual verification and rule-based systems, but these methods were limited in scalability and adaptability to large volumes of data.

One of the foundational studies in fake news detection was conducted by Shu et al. (2017), who presented a comprehensive data mining perspective on fake news. The study emphasized the importance of combining content-based analysis with social context features, such as user behavior and network propagation patterns. This work laid the groundwork for many subsequent machine learning-based approaches.

Zubiaga et al. (2018) explored the detection and resolution of rumors on social media platforms. Their research highlighted the complexity of misinformation, particularly in dynamic environments like Twitter, where information evolves rapidly. The authors proposed models that incorporate temporal and contextual information to improve classification accuracy.

Conroy et al. (2015) focused on automatic deception detection techniques. They examined linguistic cues, writing styles, and semantic patterns to differentiate between truthful and deceptive content. Their findings demonstrated that language-based features can be highly effective in identifying fake news when combined with machine learning algorithms.

Ahmed et al. (2017) introduced the use of n-gram analysis for fake news detection. By analyzing sequences of words, the study captured contextual patterns that are often missed by simple keyword-based methods. The research showed that machine learning classifiers trained on n-gram features could achieve promising results in identifying fake news articles.

Wang (2017) developed the LIAR dataset, one of the most widely used benchmark datasets for fake news detection. This dataset contains labeled short statements from political contexts and has been extensively used to evaluate various machine learning and deep learning models. The availability of such datasets has significantly accelerated research in this domain.

With the advancement of machine learning, traditional algorithms such as Logistic Regression and Naive Bayes have been widely applied for text classification tasks. These models are computationally efficient and provide good baseline performance. Logistic Regression, in particular, has been effective for binary classification problems like fake news detection due to its simplicity and interpretability.

Ensemble learning methods such as Random Forest have also been explored to improve classification performance. Random Forest combines multiple decision trees to reduce overfitting and increase robustness. Studies have shown that ensemble models generally outperform single classifiers by capturing complex patterns in data more effectively.

Gradient boosting techniques, especially Extreme Gradient Boosting (XGBoost), have gained popularity due to their high accuracy and efficiency. XGBoost uses an optimized boosting framework that iteratively improves model performance by minimizing errors. Researchers have demonstrated that XGBoost performs exceptionally well in text classification tasks, including fake news detection.

In addition to traditional machine learning methods, deep learning approaches such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have been applied to fake news detection. These models are capable of capturing sequential dependencies in text, making them suitable for analyzing complex linguistic structures. However, they require large datasets and high computational resources.

Natural Language Processing techniques play a crucial role in fake news detection systems. Feature extraction methods such as Term Frequency–Inverse Document Frequency (TF-IDF) and word embeddings are commonly used to convert textual data into numerical form. TF-IDF, in particular, is widely used due to its effectiveness in highlighting important words in a document while reducing the impact of common words.

Recent research trends focus on hybrid approaches that combine multiple models and data sources. These systems integrate content-based features with metadata, user behavior, and network information to improve detection accuracy. Additionally, explainable AI techniques are being explored to provide transparency in model predictions, helping users understand why a particular news article is classified as fake or real.

In summary, the literature indicates that fake news detection is a complex and evolving problem that requires a combination of machine learning, NLP techniques, and robust datasets. While traditional models provide a strong foundation, advanced methods such as ensemble learning and deep learning continue to enhance performance. The insights gained from these studies have significantly influenced the design and implementation of modern fake news detection systems, including the one proposed in this project.

CHAPTER–3

System Analysis:

System analysis is a crucial phase in the development of the Fake News Detection System, focusing on understanding the requirements, identifying system functionalities, and defining how different components interact. The system is designed to automatically analyze and classify news articles as real or fake using machine learning and natural language processing techniques. It involves studying user requirements, data flow, processing mechanisms, and system architecture to ensure efficient performance, scalability, and usability. The analysis also considers factors such as accuracy, response time, data handling, and user interaction through a web-based interface.

Problem Statement:

The widespread dissemination of fake news on digital platforms has created a serious challenge in maintaining information authenticity and public trust. Traditional methods of fact-checking are manual, time-consuming, and unable to handle the massive volume of online content. Existing automated approaches often lack accuracy and fail to capture contextual meaning. Therefore, there is a need for an intelligent, automated system that can efficiently analyze textual content and accurately classify news as real or fake in real time, helping users make informed decisions.

Existing System:

The existing systems for fake news detection primarily rely on manual verification or basic automated techniques such as keyword matching and rule-based filtering. Manual fact-checking involves human experts verifying the authenticity of news articles, which is highly accurate but not scalable. Some automated systems use simple algorithms like keyword detection or sentiment analysis, which are fast but often fail to understand the context and deeper meaning of the content. These systems are limited in handling large datasets and lack adaptability to evolving patterns of misinformation.

Disadvantages of Existing System:

The existing systems suffer from several limitations that reduce their effectiveness in real-world applications. Manual verification is slow, labor-intensive, and cannot keep up with the rapid spread of information. Rule-based and keyword-based systems lack contextual understanding, leading to inaccurate classifications. Many systems do not utilize advanced machine learning techniques, resulting in lower accuracy and poor generalization. Additionally, these systems often lack user-friendly interfaces and real-time prediction capabilities, making them less practical for everyday use.

Proposed System:

The proposed system is an automated Fake News Detection System that uses machine learning algorithms and natural language processing techniques to classify news articles. The system preprocesses textual data, extracts features using TF-IDF vectorization, and applies multiple classifiers such as Logistic Regression, Random Forest, and XGBoost. A Flask-based web application provides an interactive interface for users to input news articles and receive predictions. The system also implements a majority voting mechanism to improve classification reliability and provides probability scores and visualizations for better understanding.

Advantages of Proposed System:

The proposed system offers several advantages over existing approaches. It provides automated and real-time detection of fake news, significantly reducing the need for manual effort. The use of multiple machine learning models improves accuracy and robustness. The system is scalable and capable of handling large datasets efficiently. Its user-friendly web interface enhances accessibility and usability. Additionally, the inclusion of visualization and probability scores increases transparency and helps users understand prediction results more effectively. The modular design also allows for future enhancements, such as integrating deep learning models and multilingual support.

OBJECTIVES

The main objectives of the Fake News Detection System are as follows:

1. To develop an automated system that can accurately classify news articles as real or fake using machine learning techniques.
2. To apply Natural Language Processing (NLP) methods such as TF-IDF for effective feature extraction from textual data.
3. To implement multiple machine learning algorithms, including Logistic Regression, Random Forest, and XGBoost, to improve prediction accuracy and reliability.
4. To design a user-friendly web interface using Flask that allows users to input news articles and receive real-time predictions.
5. To provide comparative analysis and visualization of model performance using metrics such as accuracy, precision, recall, and F1-score.

FEASIBILITY STUDY

Feasibility study is an important step in system development that evaluates whether the proposed system is practical and achievable within the given constraints. It helps in determining the viability of the project from different perspectives such as economic, technical, and social aspects.

1. Economical Feasibility:

The proposed system is economically feasible as it requires minimal investment. The development is carried out using open-source technologies such as Python, Flask, and machine learning libraries like Scikit-learn and XGBoost, which are freely available. The hardware requirements are basic and commonly available, such as a standard computer with moderate RAM and storage. Since no expensive software licenses or specialized hardware are required, the overall cost of development and deployment is low.

2. Technical Feasibility:

The system is technically feasible as it uses well-established technologies and frameworks. Python provides strong support for machine learning and data processing, while Flask enables easy development of web applications. The use of TF-IDF for feature extraction and machine learning algorithms like Logistic Regression, Random Forest, and XGBoost ensures efficient and accurate classification. The system can be implemented and maintained with available technical resources and knowledge, making it highly practical.

3. Social Feasibility:

The proposed system is socially beneficial as it helps in reducing the spread of fake news and misinformation. By providing users with a tool to verify the authenticity of news articles, it promotes awareness and responsible information sharing. The system supports informed decision-making and contributes to maintaining social harmony by preventing the negative impacts of misleading information. It is user-friendly and accessible, making it acceptable to a wide range of users.

CHAPTER – 4

System Requirements:

System requirements define the necessary resources and conditions required for the successful development, deployment, and operation of the Fake News Detection System. These requirements include both hardware and software components, as well as functional and non-functional specifications that ensure the system performs efficiently and meets user expectations.

Software Requirements:

The software requirements for the system include the tools, programming languages, and frameworks used during development and execution:

- Operating System: Windows 7 or above / Linux / macOS
- Programming Language: Python 3.x
- Web Framework: Flask
- Frontend Technologies: HTML, CSS, JavaScript
- Database: MySQL
- Libraries and Packages:
    - Scikit-learn (for machine learning models)
    - Pandas and NumPy (for data processing)
    - Matplotlib (for visualization)
    - Joblib (for model saving and loading)
    - XGBoost (for advanced boosting algorithm)
- Web Browser: Google Chrome, Mozilla Firefox

Hardware Requirements:

The hardware requirements specify the minimum system configuration needed to run the application:

- Processor: Intel Pentium IV or higher
- RAM: Minimum 4 GB
- Storage: At least 20 GB free disk space
- Input Devices: Keyboard and Mouse
- Display: Monitor with minimum 1024 × 768 resolution

Functional Requirements:

Functional requirements describe the core functionalities and operations of the system:

- The system should allow users to register and log in securely.
- The system should provide role-based access (Admin and User).
- The admin should be able to upload datasets for model training.
- The system should preprocess textual data (cleaning and feature extraction).
- The system should train multiple machine learning models (Logistic Regression, Random Forest, XGBoost).
- The system should evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.
- The system should allow users to input news articles for prediction.
- The system should classify news as real or fake in real time.
- The system should display prediction results along with probability scores.
- The system should generate graphical visualizations of model performance and predictions.

Non-Functional Requirements:

Non-functional requirements define the quality attributes and performance characteristics of the system:

- **Performance:** The system should provide fast and accurate predictions with minimal response time.

- **Scalability:** The system should handle increasing amounts of data and users efficiently.
- **Security:** User data and credentials should be protected using secure authentication mechanisms.
- **Usability:** The interface should be user-friendly, intuitive, and easy to navigate.
- **Reliability:** The system should function consistently without failures or crashes.
- **Maintainability:** The system should be modular and easy to update or enhance.
- **Compatibility:** The system should work across different operating systems and web browsers.
- **Availability:** The system should be accessible whenever users need it without significant downtime.

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SYSTEM DESIGN

System Architecture:

The Fake News Detection System follows a three-tier architecture that ensures modularity, scalability, and efficient data processing. The architecture is divided into three main layers: Presentation Layer, Application Layer, and Data Layer.

The Presentation Layer consists of a web-based user interface developed using Flask along with HTML, CSS, and JavaScript. It allows users to interact with the system through features such as registration, login, dataset upload, and news prediction. This layer is responsible for collecting user input and displaying results in a user-friendly format.

The Application Layer handles the core logic of the system. It includes data preprocessing, feature extraction using TF-IDF, model training, and prediction processes. Machine learning algorithms such as Logistic Regression, Random Forest, and XGBoost are implemented in this layer. It also manages user sessions, authentication, and communication between the frontend and backend.

The Data Layer is responsible for storing and managing data. It uses a MySQL database to store user information, dataset details, and model performance metrics. Additionally, trained machine learning models and vectorizers are saved as files on the server for future use.

Modules:

The system is divided into several functional modules to ensure efficient operation and easy maintenance:

1. **User Management Module:**
   This module handles user registration, login, and authentication. It manages user roles (Admin and User) and ensures secure access to system features.

2. **Dataset Upload Module:**
   This module allows administrators to upload datasets in CSV or Excel format. It validates and stores the dataset for further processing.

3. **Data Preprocessing Module:**
   This module cleans the raw text data by removing stopwords, punctuation, and irrelevant characters. It prepares the data for feature extraction.

4. **Feature Extraction Module:**
   This module converts textual data into numerical form using TF-IDF vectorization, enabling machine learning models to process the data.

5. **Model Training Module:**
   This module trains multiple machine learning models such as Logistic Regression, Random Forest, and XGBoost. It evaluates their performance using standard metrics.

6. **Prediction Module:**
   This module allows users to input news articles and predicts whether the news is real or fake. It uses trained models and applies a majority voting mechanism.

7. **Visualization Module:**
   This module generates graphical representations such as bar charts for model performance and prediction probabilities.

8. **System Interface Module:**
   This module integrates all components and provides a smooth interaction between the user and the system through the web interface.

UML Diagrams

Unified Modeling Language (UML) diagrams are used to visually represent the structure and behavior of the system.

1. **Use Case Diagram:**
   The Use Case Diagram represents the interaction between users (Admin and User) and the system. The Admin can upload datasets, train models, and view performance metrics, while the User can register, log in, and predict news authenticity.

2. **Class Diagram:**
   The Class Diagram shows the structure of the system by defining classes such as User, Dataset, Model, and Prediction. It includes attributes (e.g., username, password, dataset name) and methods (e.g., login(), upload(), train(), predict()) along with relationships between classes.

3. **Sequence Diagram:**
   The Sequence Diagram illustrates the flow of interactions over time. It shows how a

user inputs news text, the system processes it through preprocessing and prediction modules, and finally returns the result.

4. **Activity Diagram:**
   The Activity Diagram represents the workflow of the system. It includes steps such as user login, dataset upload, model training, and prediction, along with decision points and process flow.

5. **Component Diagram:**
   The Component Diagram shows how different components such as the web interface, backend logic, machine learning models, and database interact with each other.

6. **Deployment Diagram:**
   The Deployment Diagram illustrates the physical deployment of the system, including the client machine (user browser), web server (Flask application), and database server (MySQL).

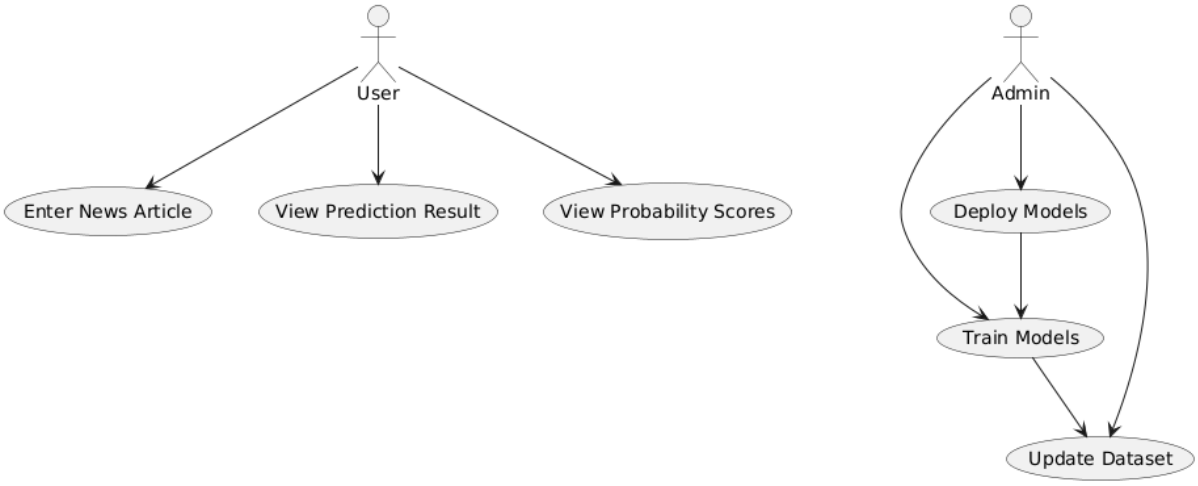
Use Case Diagram

Purpose:

- Represents the interactions between users (actors) and the system.
- Shows the system's functionality from a high-level perspective.

Components:

- **Actors**: Represent users or other systems interacting with the system.
- **Use Cases**: Represent system functionalities or services.
- **Relationships**: Includes associations, generalizations, and dependencies between actors and use cases.



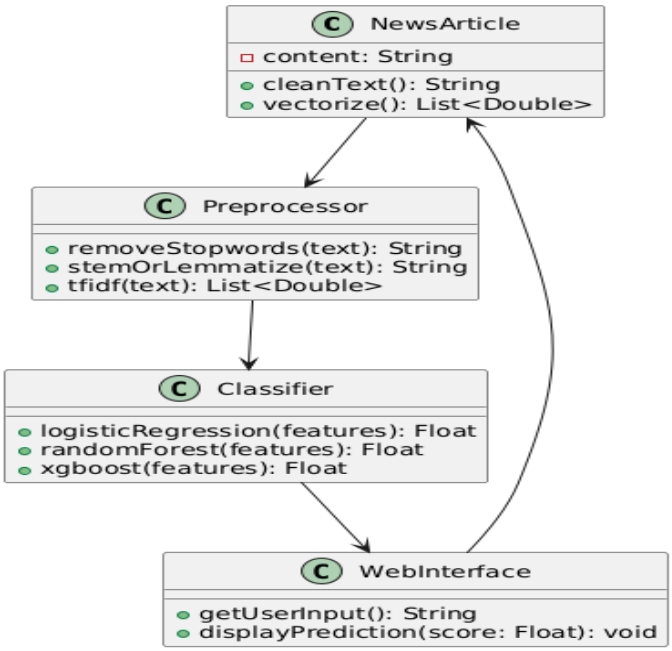
Class Diagram

Purpose:

- Defines the system's structure by showing classes, attributes, methods, and relationships.

Components:

- **Classes**: Represent objects in the system, with attributes and methods.
- **Relationships**: Includes associations, dependencies, generalization (inheritance), and aggregation/composition.



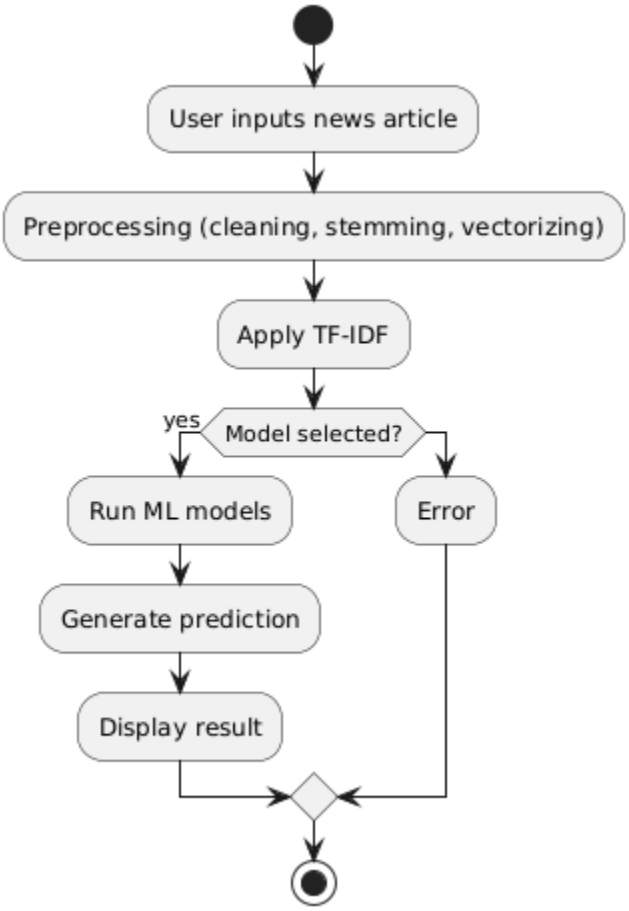
Activity Diagram

Purpose:

- Represents workflow or process flow in a system.

Components:

- **Start/End Node**: Represents the beginning and end of the flow.
- **Actions/Activities**: Represent steps in a process.
- **Decision Nodes**: Represent conditional branches.
- **Arrows**: Indicate the flow of actions.



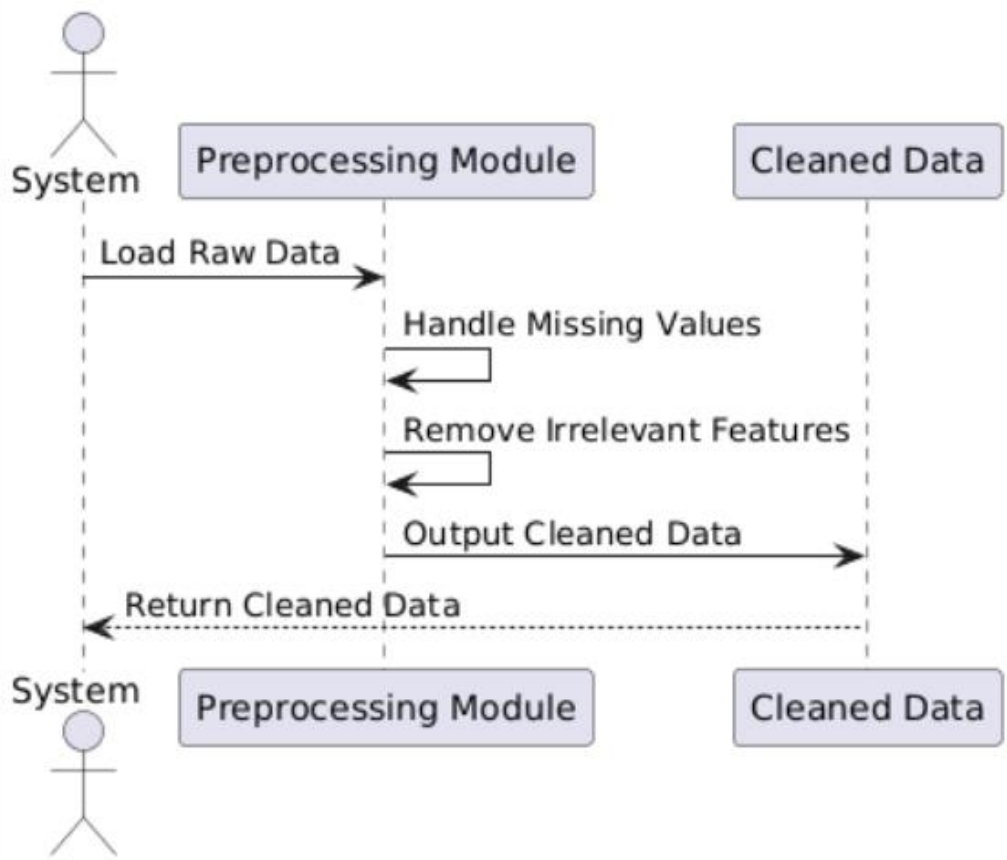
Sequence Diagram

Purpose:

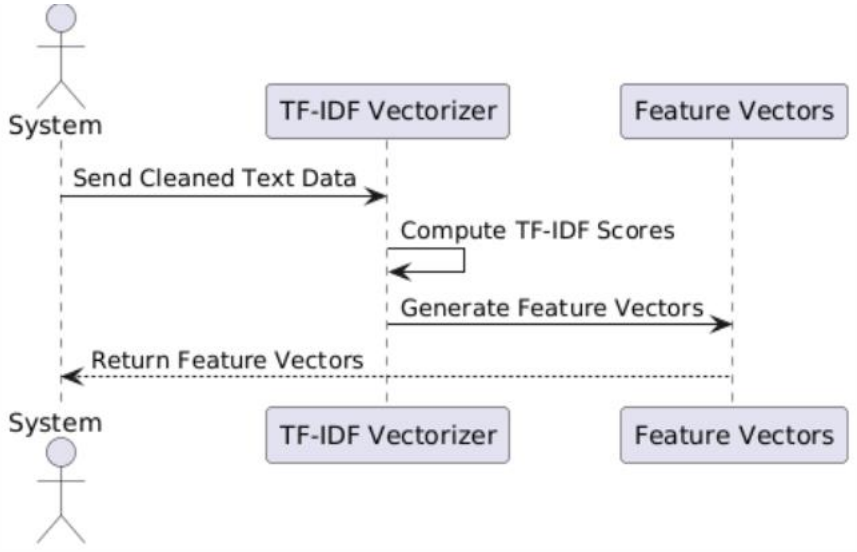
- Shows interactions between objects over time.

Components:

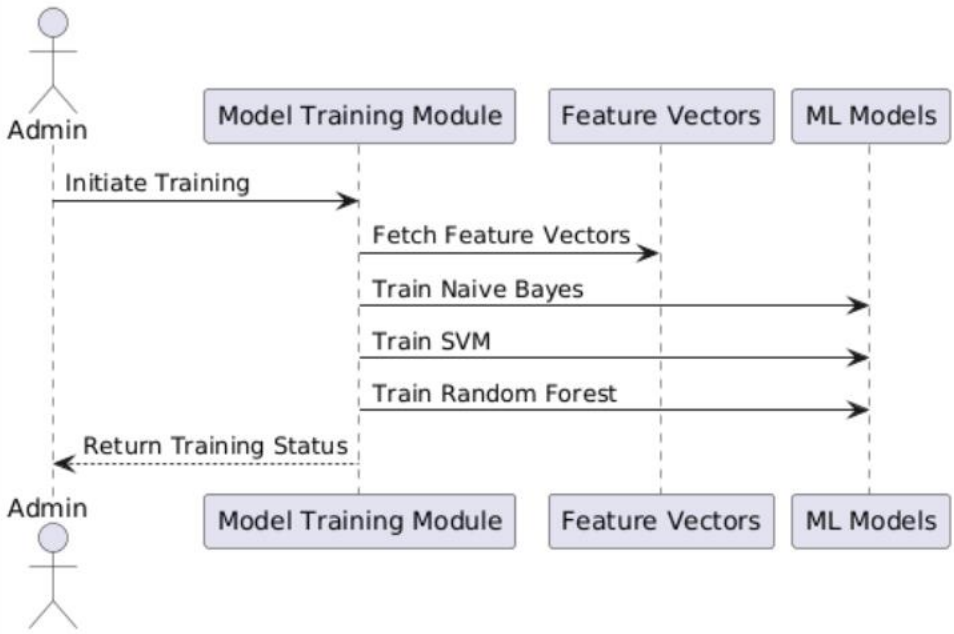
- **Actors**: Represent external entities (users or systems).
- **Objects**: Represent system components.
- **Lifelines**: Show the lifespan of an object.
- **Messages**: Represent interactions between objects.
- **Admin: Data Pre-processing Module**



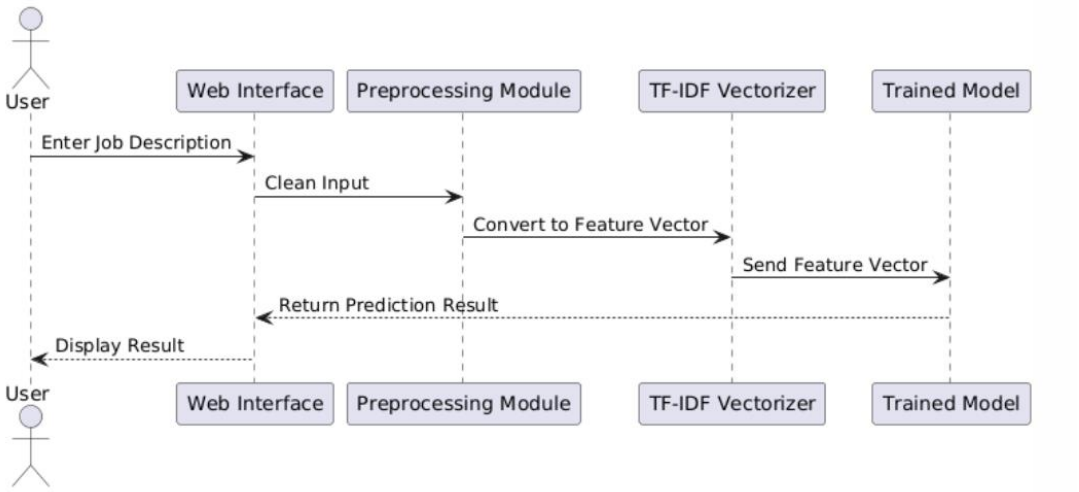
Admin:
Feature Extraction Module



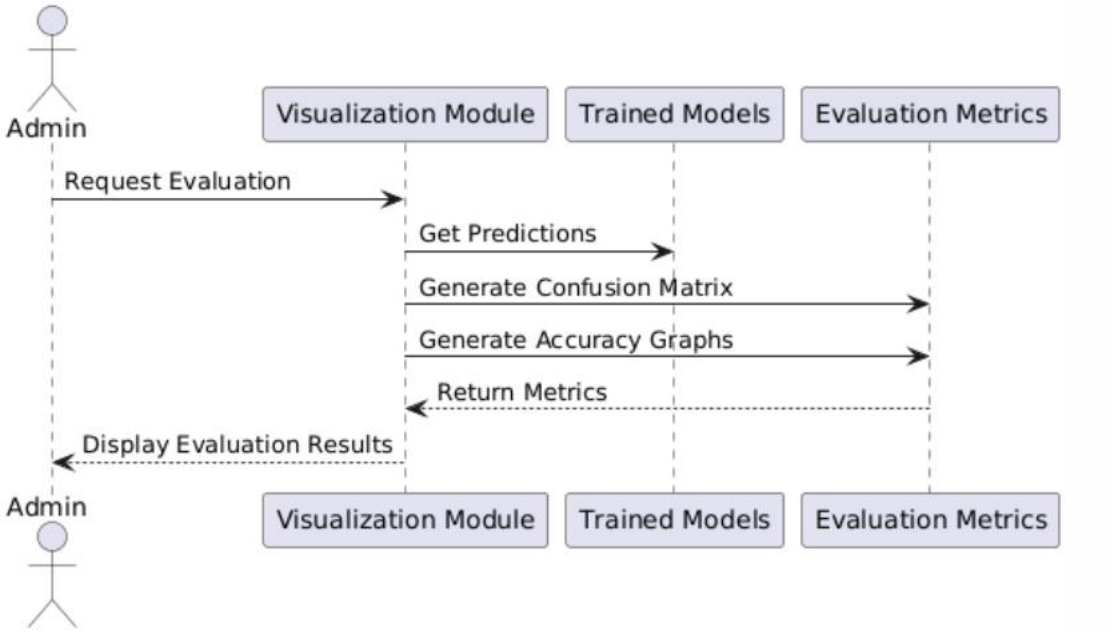
Admin: Model Training Module



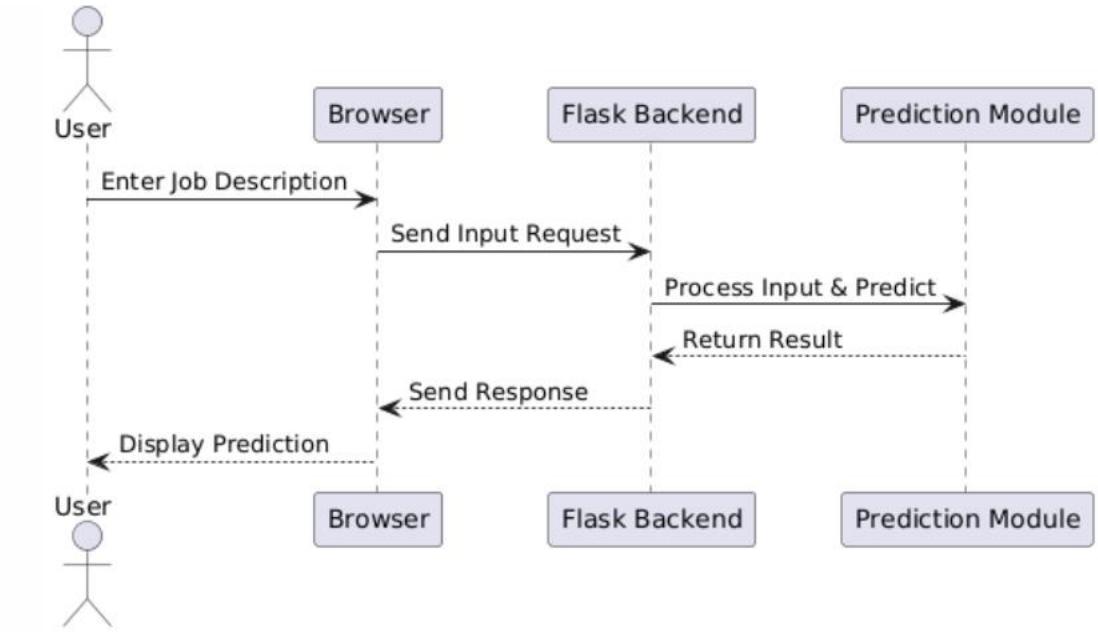
User: Prediction Module



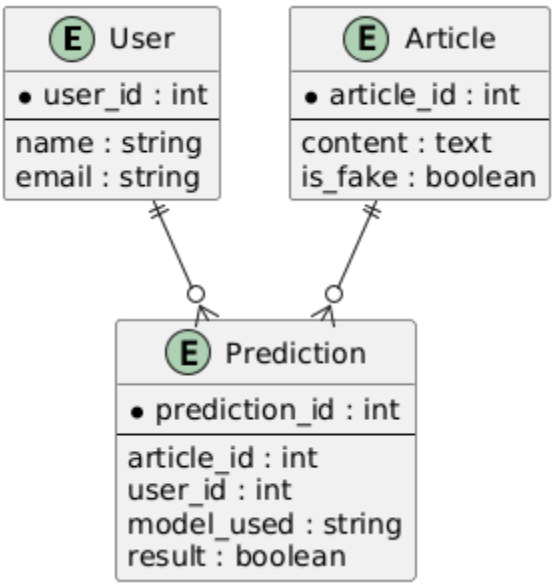
User: Visualization & Analysis Module



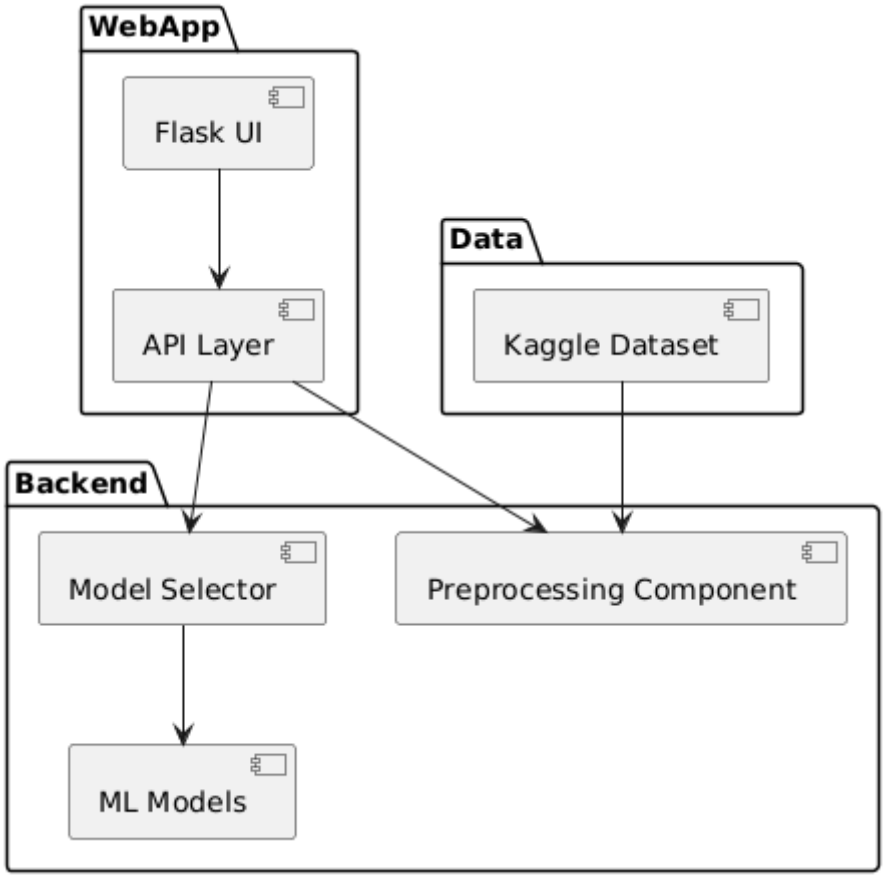
Web Interface Module (Flask)



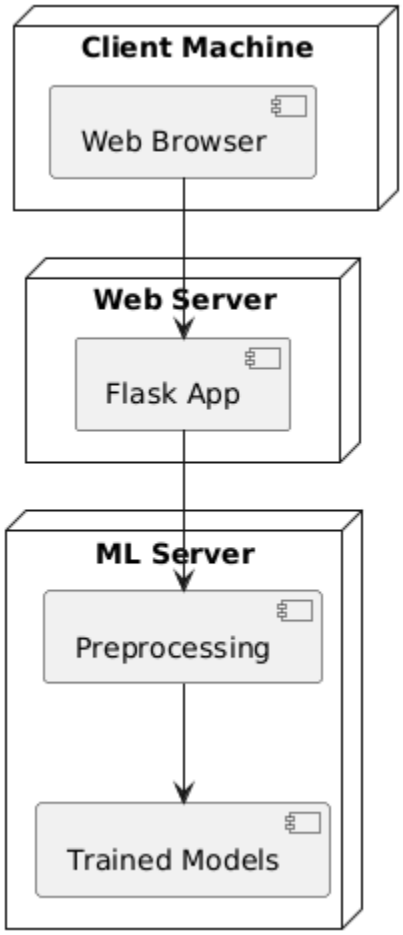
ENTITY RELATIONSHIP DIAGRAM



COMPONENT DIAGRAM USING



DEPLOYMENT DIAGRAM



CHAPTER – 6

METHODOLOGY AND IMPLEMENTATION

The Fake News Detection System is developed using a structured methodology that combines machine learning techniques with natural language processing (NLP) to classify news articles as real or fake. The implementation follows a systematic workflow consisting of data collection, preprocessing, feature extraction, model training, evaluation, and deployment through a web application.

The first step in the methodology is data collection, where a labeled dataset of news articles is obtained from reliable sources such as Kaggle. The dataset contains textual content along with labels indicating whether the news is real or fake. This data serves as the foundation for training machine learning models.

The next step is data preprocessing, which involves cleaning the raw text data. This includes removing stopwords, punctuation, special characters, and irrelevant symbols. The text is also converted to lowercase to maintain consistency. Preprocessing ensures that the data is in a suitable format for further analysis and improves the performance of machine learning algorithms.

Feature extraction is performed using the Term Frequency–Inverse Document Frequency (TF-IDF) technique. TF-IDF converts textual data into numerical vectors by measuring the importance of words in a document relative to the entire dataset. This transformation allows machine learning models to process and analyze textual information effectively.

Following feature extraction, the dataset is divided into training and testing sets. The training set is used to build the models, while the testing set is used to evaluate their performance. This step is important to ensure that the models generalize well to unseen data and do not overfit.

Multiple machine learning algorithms are then applied for classification. Logistic Regression is used as a baseline model due to its simplicity and efficiency in binary classification. Random Forest is used as an ensemble method that combines multiple decision trees to improve accuracy

and reduce overfitting. XGBoost, a gradient boosting algorithm, is used to enhance performance by optimizing the learning process through iterative improvements.

After training, the models are evaluated using performance metrics such as accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of model performance and help in comparing different algorithms. The trained models and TF-IDF vectorizer are then saved using joblib for future use.

The implementation also includes a Flask-based web application that serves as the user interface. The application allows users to register, log in, and input news articles for prediction. When a user submits text, it is processed using the saved vectorizer, and predictions are generated using the trained models. A majority voting mechanism is applied to determine the final classification result. The system also displays probability scores and visualizations to enhance user understanding.

Algorithms Used in the Project:

The system utilizes the following machine learning algorithms:

1. **Logistic Regression:**

   Logistic Regression is a supervised learning algorithm used for binary classification problems. It estimates the probability that a given input belongs to a particular class using a logistic function. It is efficient, easy to implement, and performs well on linearly separable data.

2. **Random Forest:**

   Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the majority class as the prediction. It reduces overfitting and improves accuracy by combining the results of multiple trees.

3. **XGBoost (Extreme Gradient Boosting):**

   XGBoost is an advanced ensemble algorithm based on gradient boosting. It builds models sequentially and optimizes performance by minimizing errors. It is known for its high accuracy, speed, and efficiency in handling large datasets.

4. **TF-IDF Vectorization:**

   TF-IDF is a feature extraction technique used to convert textual data into numerical form. It assigns weights to words based on their importance in a document relative to the dataset, improving the effectiveness of machine learning models.

Dataset Used in the Project:

The dataset used in this project is obtained from Kaggle, which contains labeled news articles categorized as real or fake. The dataset includes two main columns: the text of the news article and its corresponding label.

The dataset undergoes preprocessing to remove noise and ensure consistency. The labels are encoded into numerical values, where “fake” is represented as 0 and “real” as 1. The cleaned dataset is then used for training and testing machine learning models.

The quality and size of the dataset play a crucial role in determining the accuracy of the system. A well-balanced dataset with sufficient examples of both real and fake news helps in building a reliable and unbiased model.

CHAPTER – 7

SOFTWARE INTRODUCTION AND TESTING

The Fake News Detection System is developed using modern software technologies that support machine learning, data processing, and web application development. The primary programming language used is Python, which is widely recognized for its simplicity, flexibility, and extensive support for data science and machine learning applications.

The system utilizes the Flask framework to develop a lightweight and efficient web application. Flask enables seamless integration between the frontend and backend, allowing users to interact with the system through a browser interface. HTML, CSS, and JavaScript are used to design responsive and user-friendly web pages.

For data processing and machine learning tasks, libraries such as Pandas and NumPy are used to handle datasets and perform numerical computations. Scikit-learn provides implementations of machine learning algorithms like Logistic Regression and Random Forest, while XGBoost is used for advanced boosting techniques. TF-IDF vectorization is applied for feature extraction from textual data.

Matplotlib is used for generating graphical visualizations of model performance and prediction results. Joblib is used for saving and loading trained models efficiently. MySQL is used as the database to store user information, dataset details, and model performance metrics.

Software Used in the Project:

- Python 3.x
- Flask Framework
- MySQL Database

- Scikit-learn Library
- Pandas and NumPy
- Matplotlib
- XGBoost
- Joblib
- HTML, CSS, JavaScript
- Web Browsers (Chrome, Firefox)

Software Testing:

Software testing is performed to ensure that the system functions correctly, meets user requirements, and produces accurate results. Different types of testing are conducted to validate the performance and reliability of the system.

1.Unit Testing:

Each module of the system, such as data preprocessing, model training, and prediction, is tested individually to ensure correct functionality.

2. Integration Testing:

The interaction between modules, such as dataset upload, model training, and prediction, is tested to ensure smooth data flow and communication.

3. Functional Testing:

All functionalities, including user registration, login, dataset upload, model training, and prediction, are tested against expected outputs.

4. Performance Testing:

The system is tested for response time and efficiency, especially during model training and real-time prediction.

5. Security Testing:

User authentication and role-based access control are tested to ensure secure access to system features.

Sample Source Code:

The following code snippets represent key parts of the implementation from the project:

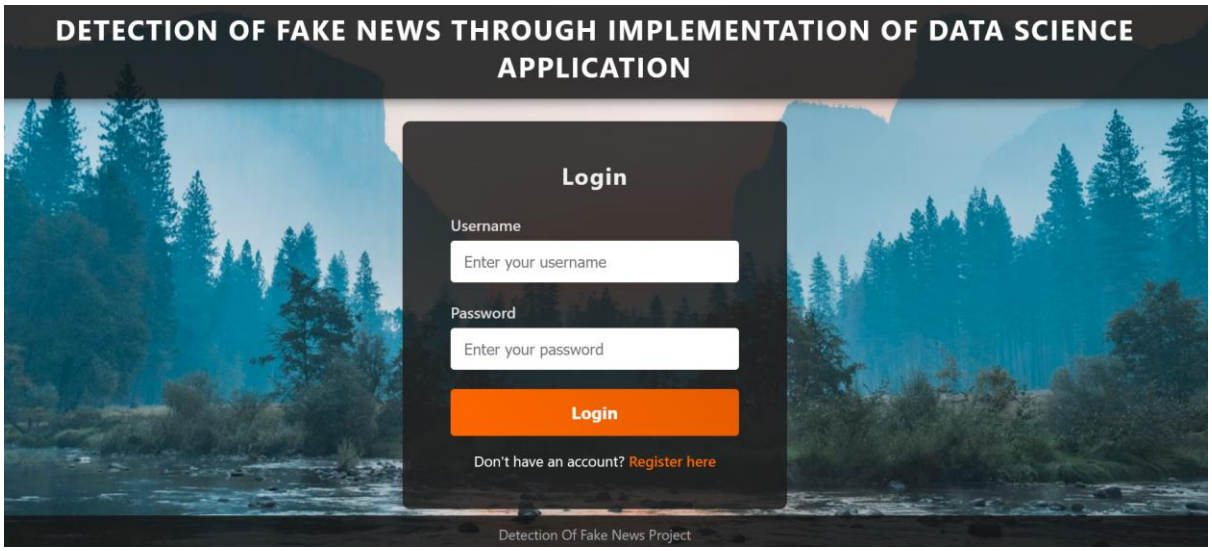
```
# Importing required libraries
from flask import Flask, render_template, request, redirect, url_for, session, flash
from flask_mysqldb import MySQL
import MySQLdb.cursors
import pandas as pd
import joblib
import os
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
# Flask App Initialization
app = Flask(__name__)
app.secret_key = 'your_secret_key'

# MySQL Configuration
app.config['MYSQL_HOST'] = 'localhost'
app.config['MYSQL_USER'] = 'root'
app.config['MYSQL_PASSWORD'] = 'root'
app.config['MYSQL_DB'] = 'fake_news_db'

mysql = MySQL(app)
# User Registration
@app.route('/register', methods=['GET', 'POST'])
def register():
```

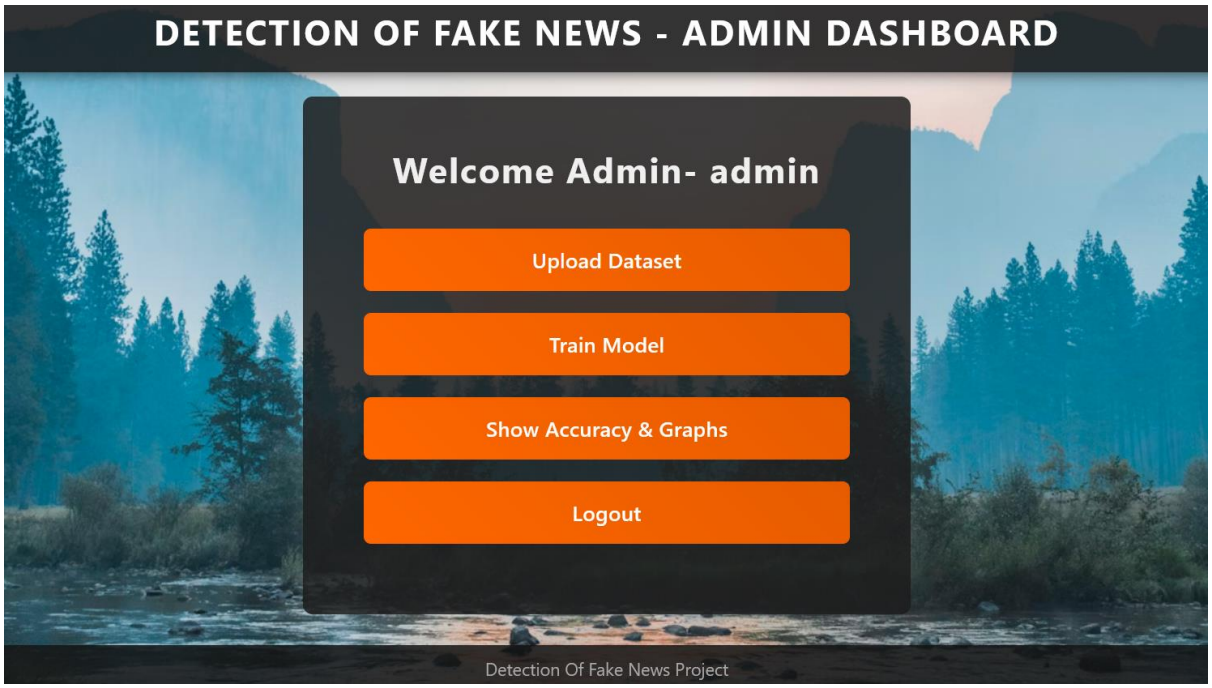
```
    if request.method == 'POST':
        username = request.form['username']
        email = request.form['email']
        password = request.form['password']

        cursor = mysql.connection.cursor()
        cursor.execute('INSERT INTO users (username, email, password) VALUES (%s,%s,%s)',
                       (username, email, password))
        mysql.connection.commit()
        return redirect(url_for('login'))
    return render_template('register.html')
# User Login
@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        username = request.form['username']
        password = request.form['password']

        cursor = mysql.connection.cursor(MySQLdb.cursors.DictCursor)
        cursor.execute('SELECT * FROM users WHERE username=%s AND password=%s',
                       (username, password))
        user = cursor.fetchone()
        if user:
            session['loggedin'] = True
            return redirect(url_for('admin_dashboard'))
    return render_template('login.html')
# Dataset Upload
@app.route('/admin/upload_dataset', methods=['POST'])
def upload_dataset():
    file = request.files['dataset']
    filepath = os.path.join('datasets', file.filename)
```

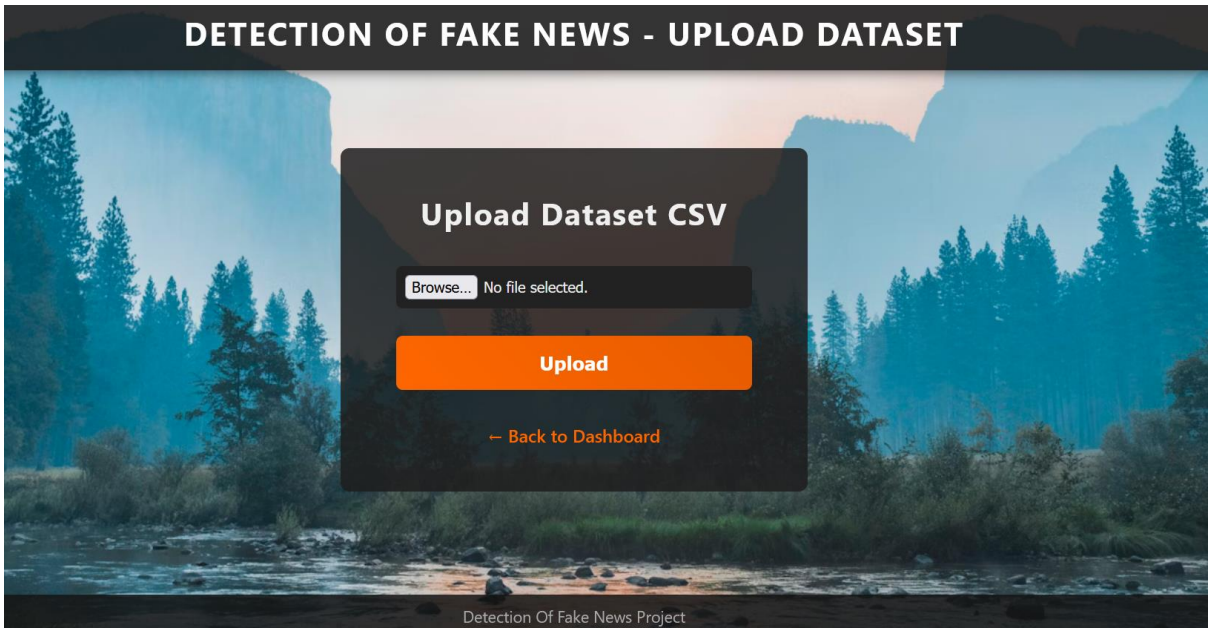
```
    file.save(filepath)
    return "Dataset uploaded successfully"
# Model Training
def train_models(texts, labels):
    vectorizer = TfidfVectorizer(max_features=5000)
    X = vectorizer.fit_transform(texts)

    lr = LogisticRegression().fit(X, labels)
    rf = RandomForestClassifier().fit(X, labels)
    xgb = XGBClassifier().fit(X, labels)

    joblib.dump(lr, 'models/lr.pkl')
    joblib.dump(rf, 'models/rf.pkl')
    joblib.dump(xgb, 'models/xgb.pkl')
# Prediction
def predict_news(text):
    vectorizer = joblib.load('models/tfidf_vectorizer.pkl')
    model = joblib.load('models/lr.pkl')

    X = vectorizer.transform([text])
    prediction = model.predict(X)
    return "Real News" if prediction[0] == 1 else "Fake News"
```

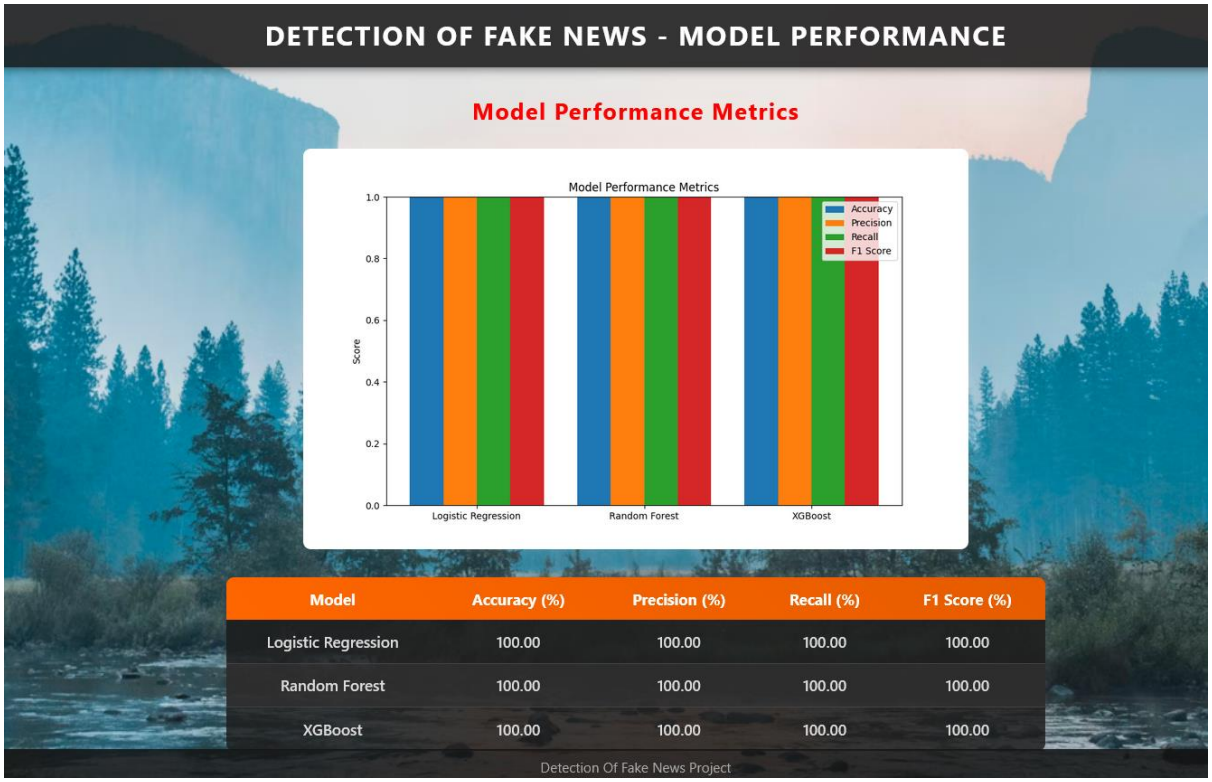
CHAPTER -8

SCREEN SHOTS AND RESULTS

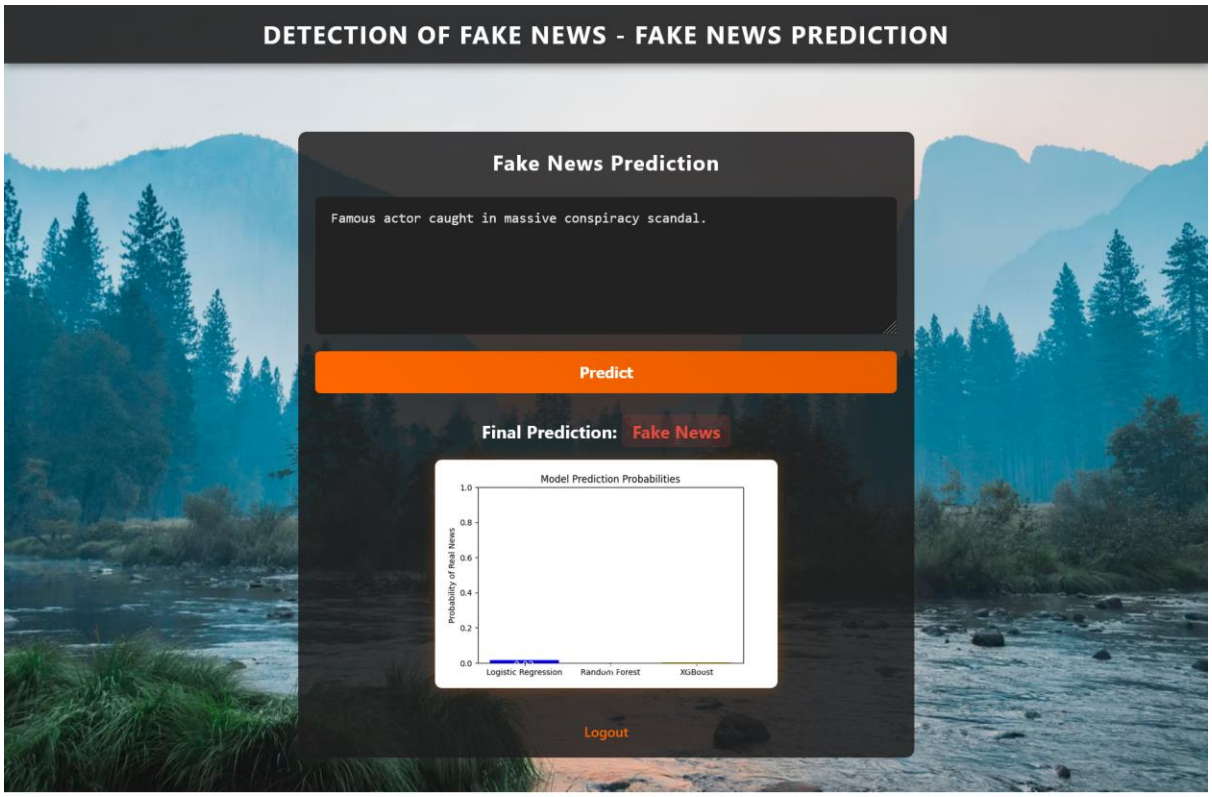


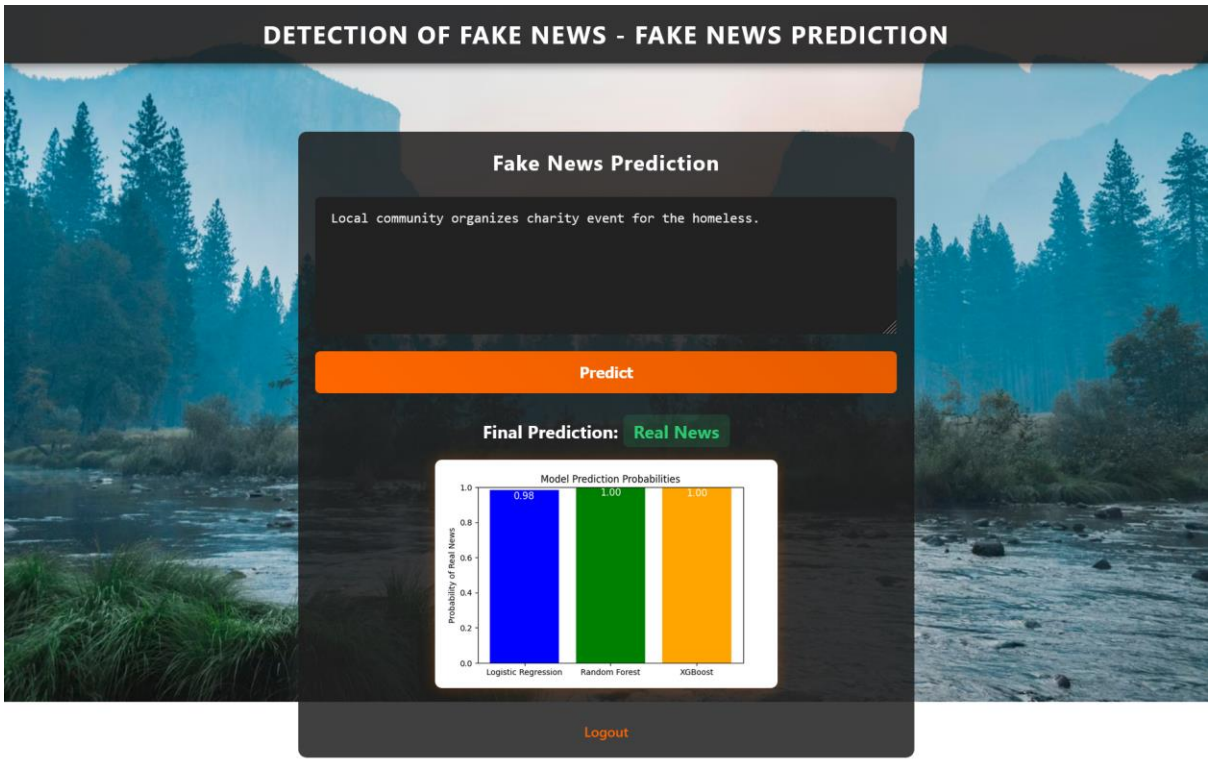






Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Logistic Regression	100.00	100.00	100.00	100.00
Random Forest	100.00	100.00	100.00	100.00
XGBoost	100.00	100.00	100.00	100.00


DETECTION OF FAKE NEWS - FAKE NEWS PREDICTION
Fake News Prediction
Famous actor caught in massive conspiracy scandal.
Predict
Final Prediction: Fake News
Model Prediction Probabilities
Probability of Real News
1.0
0.8
0.6
0.4
0.2
0.0
Logistic Regression
Random Forest
XGBoost
Logout


DETECTION OF FAKE NEWS - FAKE NEWS PREDICTION
Fake News Prediction
Local community organizes charity event for the homeless.
Predict
Final Prediction: Real News
Model Prediction Probabilities
Probability of Real News
0.98
1.00
1.00
Logistic Regression
Random Forest
XGBoost
Logout

9.CONCLUSION

The Fake News Detection System developed in this project successfully demonstrates the application of machine learning and natural language processing techniques to address the growing problem of misinformation in the digital age. With the rapid increase in online content, the need for automated and reliable systems to verify the authenticity of news has become essential.

In this project, a comprehensive approach was implemented that includes data preprocessing, feature extraction using TF-IDF, and classification using multiple machine learning algorithms such as Logistic Regression, Random Forest, and XGBoost. These models were trained and evaluated using standard performance metrics including accuracy, precision, recall, and F1-score, ensuring a reliable assessment of their effectiveness.

The integration of multiple models along with a majority voting mechanism improved the overall accuracy and robustness of the system. The use of a Flask-based web application provided an interactive and user-friendly interface, allowing users to easily input news articles and obtain real-time predictions along with probability scores and visualizations.

The system effectively achieves its objectives by providing an automated, scalable, and efficient solution for fake news detection. It reduces dependency on manual verification and enables users to make informed decisions based on data-driven predictions. The modular design of the system also ensures flexibility for future enhancements and integration of advanced technologies.

However, like any system, it has certain limitations, such as dependency on the quality of the dataset and focus on textual content only. Despite these limitations, the project lays a strong foundation for further research and development in the field of fake news detection.

In conclusion, this project highlights the potential of machine learning and NLP in solving real-world problems and contributes towards promoting reliable information sharing in society. With future improvements such as deep learning integration, multilingual support, and real-time verification, the system can become even more powerful and impactful.

FUTURE SCOPE

The Fake News Detection System developed in this project provides a strong foundation for identifying misleading information using machine learning and natural language processing techniques. However, there are several opportunities for further enhancement and improvement to increase its effectiveness and real-world applicability.

One of the major future enhancements is the integration of deep learning models such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and transformer-based models like BERT. These models can better understand contextual and semantic relationships in text, leading to improved accuracy in fake news detection.

Another important improvement is the incorporation of multilingual support. Currently, the system primarily focuses on English-language news articles. Extending the system to handle multiple languages will make it more useful and accessible to a wider audience across different regions.

The system can also be enhanced by integrating real-time fact-checking APIs and external knowledge sources. This would allow the system to verify information against trusted databases and provide more reliable results rather than relying solely on trained models.

In addition, the inclusion of multimedia analysis such as image and video verification can significantly improve the system. Fake news often includes manipulated images or videos, and analyzing such content using computer vision techniques can provide a more comprehensive detection mechanism.

Another potential improvement is the deployment of the system as a mobile application or browser extension. This would allow users to quickly verify news articles while browsing social media or websites, making the system more practical for everyday use.

The system can also benefit from continuous learning by updating models with new data over time. This will help the system adapt to evolving patterns of fake news and improve its performance.

Finally, incorporating explainable AI techniques can enhance transparency by providing reasons for predictions. This will help users understand why a particular news article is classified as fake or real, thereby increasing trust in the system.

In conclusion, the proposed system has significant potential for future expansion and improvement, making it more jgaccurate, scalable, and impactful in combating the spread of fake news.

10: REFERENCES

1. Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake News Detection on Social Media: A Data Mining Perspective. *ACM SIGKDD Explorations Newsletter,* 19(1), 22-36. https://doi.org/10.1145/3137597.3137600

2. Zubiaga, A., Aker, A., Bontcheva, K., Liakata, M., & Procter, R. (2018). Detection and Resolution of Rumours in Social Media: A Survey. *ACM Computing Surveys*, 51(2), 32. https://doi.org/10.1145/3176213

3. Ahmed, H., Traore, I., & Saad, S. (2017). Detection of Online Fake News Using N-gram Analysis and Machine Learning Techniques. *Proceedings of the 10th International Conference on Social Computing and Social Media*, 127-138.

4. Conroy, N. J., Rubin, V. L., & Chen, Y. (2015). Automatic Deception Detection: Methods for Finding Fake News. *Proceedings of the 78th ASIS&T Annual Meeting*, 82.

5. Wang, W. Y. (2017). "Liar, Liar Pants on Fire": A New Benchmark Dataset for Fake News Detection. *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 422-426.

6. Python Software Foundation. (2023). *Python Language Reference, version 3.11*. https://www.python.org/

7. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, É. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.

8. Flask Documentation. (2023). *Flask Web Framework*. https://flask.palletsprojects.com/